# 1. Project overview

KrishiConnect is an agricultural trade, knowledge-sharing and market-insights platform. It connects farmers, sellers and buyers so that produce can be listed and discovered directly, price decisions can be made with transparent mandi data, and farmers can learn from each other in a community forum.

This document presents the working application through screenshots taken from the running system at localhost:8000. Each screenshot is followed by a short description of what it demonstrates.

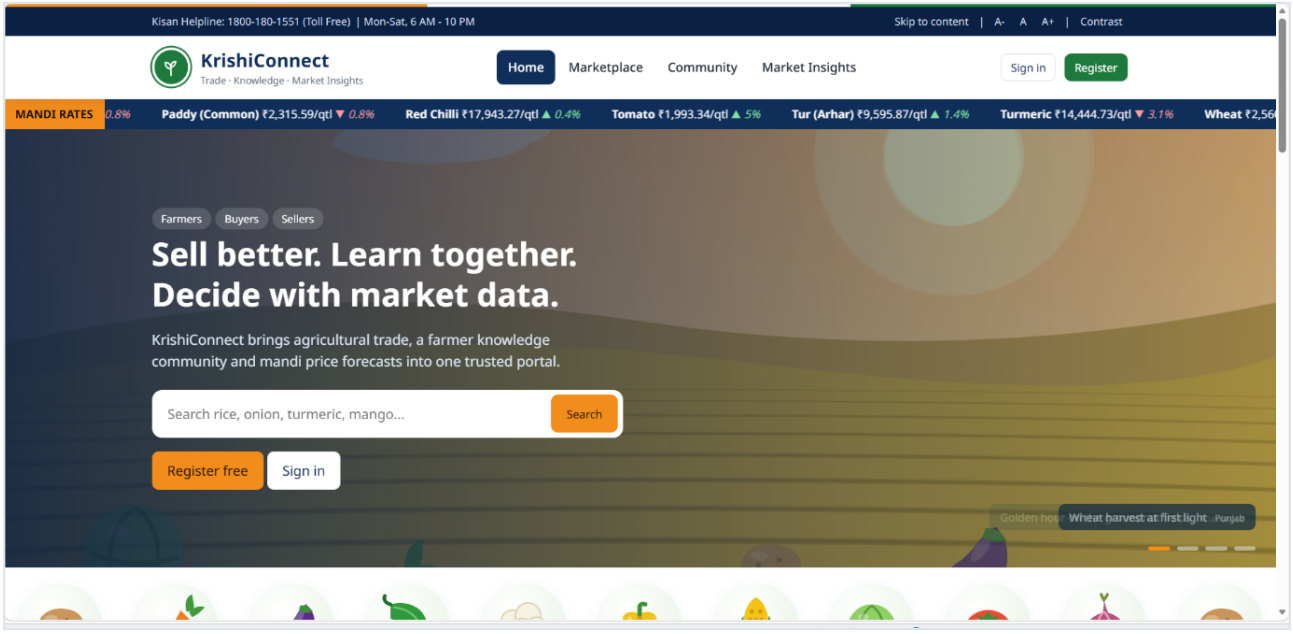
## Technology stack

| | |
|---|---|
| **Backend** | FastAPI, SQLAlchemy, PostgreSQL (single database) |
| **Security** | JWT authentication, bcrypt password hashing, role-based access (farmer, seller, buyer, admin) |
| **Analytics** | ARIMA price forecasting with statsmodels; 7-day moving average |
| **Frontend** | Vanilla JavaScript single-page application with hash routing; Leaflet / OpenStreetMap for locations |
| **Extras** | QR-code product traceability, notifications, accessibility controls (text size, contrast, skip link) |

## Screens covered

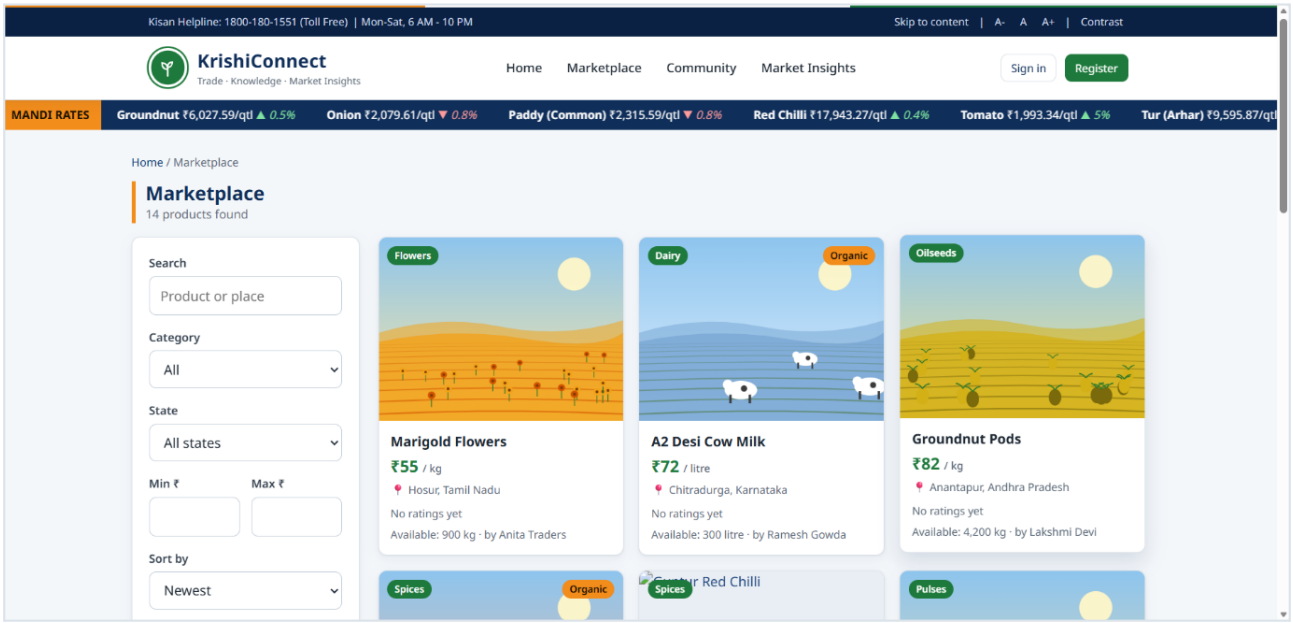
Home page, Marketplace, List a product, Market Insights (summary and price chart with forecast), and the Farmer Community in signed-in and signed-out views.

## Figure 1. Home page



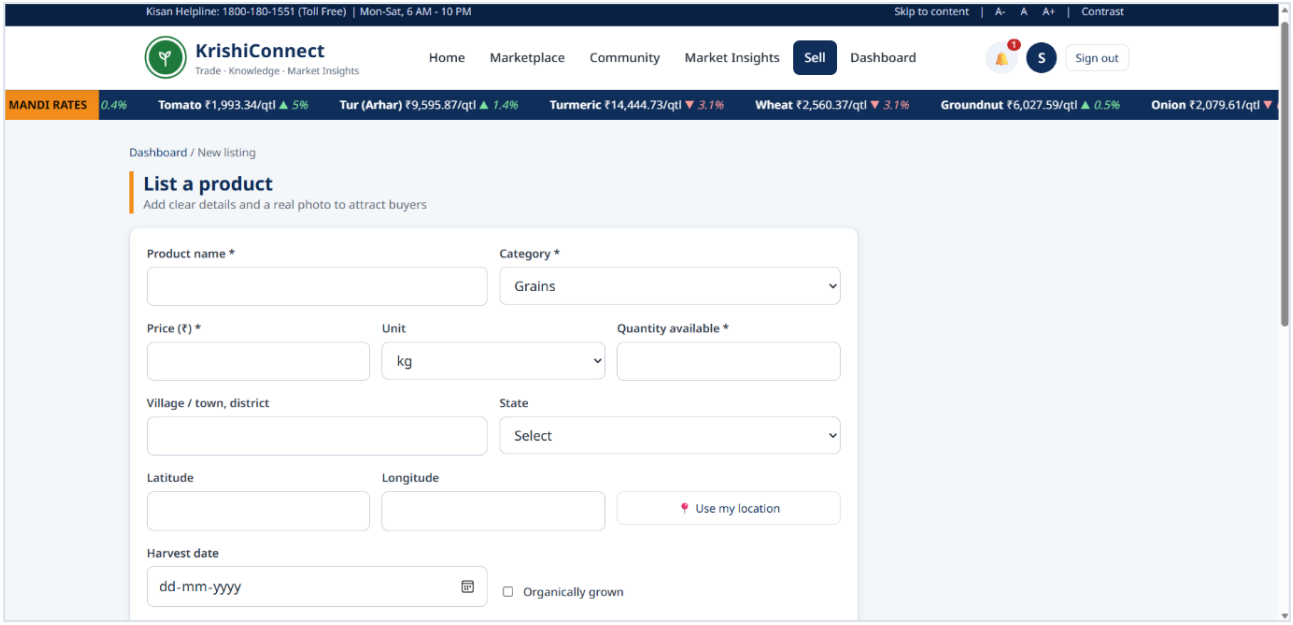

The landing page opens with the Kisan helpline bar, accessibility controls (text size and contrast), the main navigation and a live mandi-rates ticker. The hero section carries the headline, a product search box and Register / Sign in actions over a rotating farm-photo slider, with a strip of vegetable categories beginning below it.

## Figure 2. Marketplace



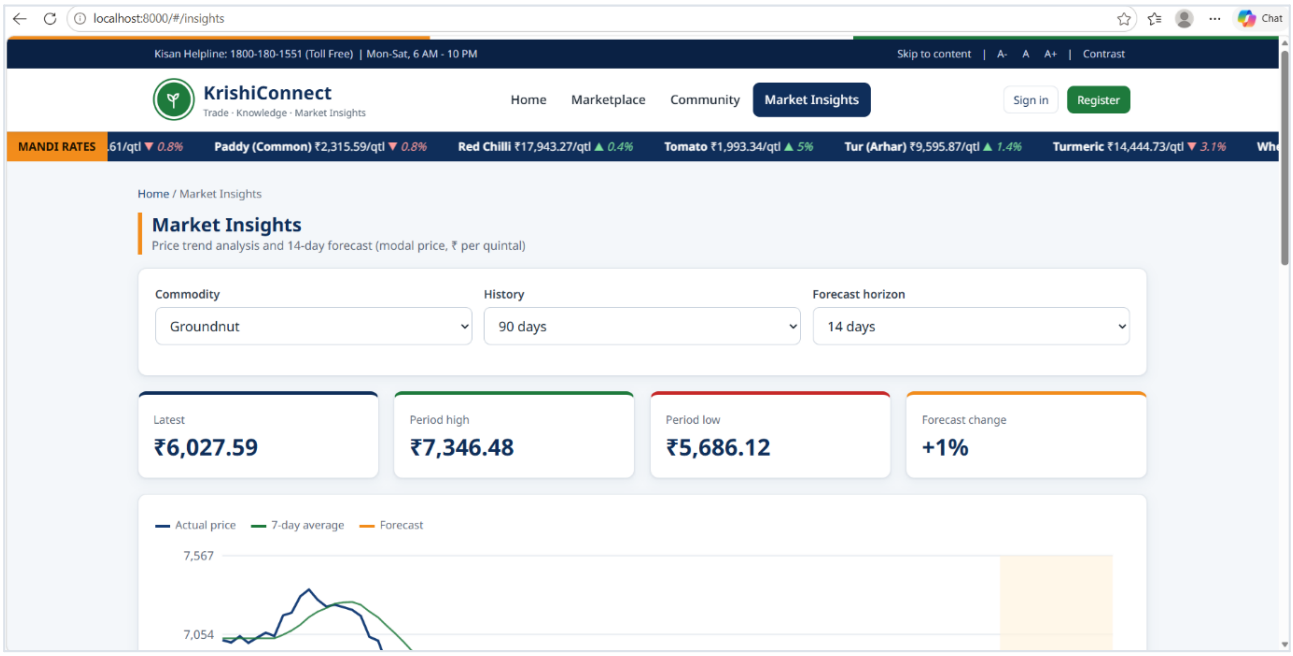

The marketplace lists 14 products with a filter panel (keyword, category, state, minimum and maximum price, sort order). Each card shows category and organic badges, price per unit, location, rating status, available quantity and the seller.

## Figure 3. List a product (seller view)



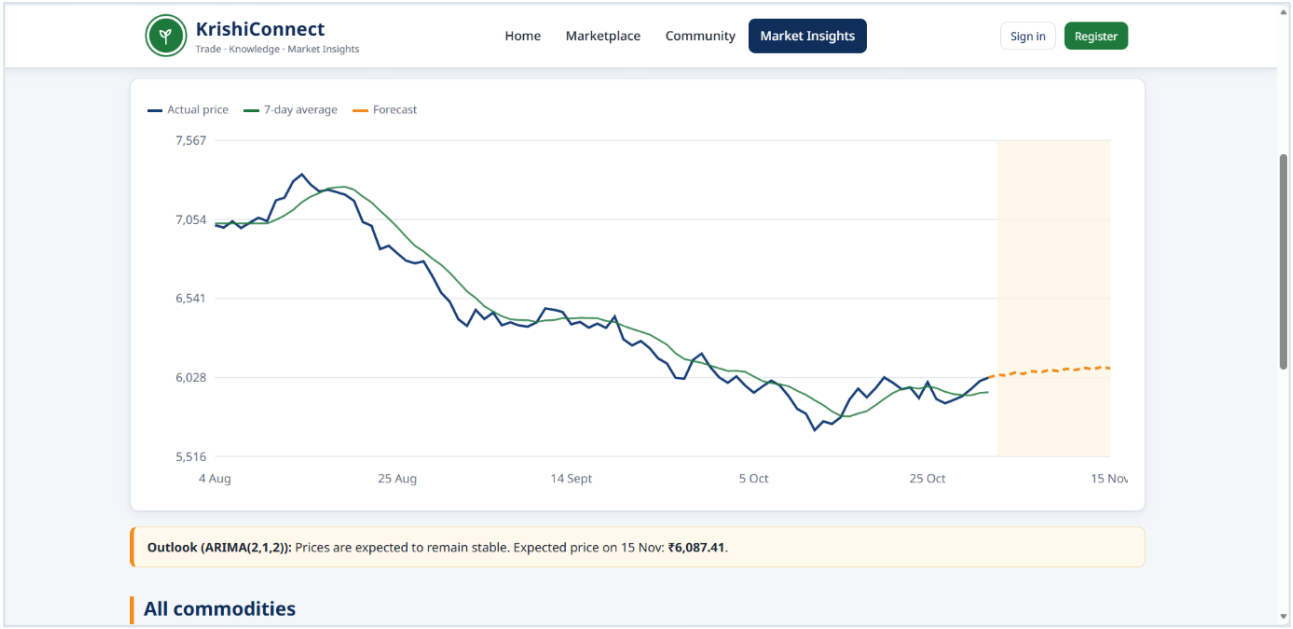

A signed-in seller adds a listing with product name, category, price, unit, quantity, village / district, state, latitude and longitude (with a Use my location button), harvest date and an organic flag. The header shows the notification bell and the Sell and Dashboard links for this role.

## Figure 4. Market Insights - summary



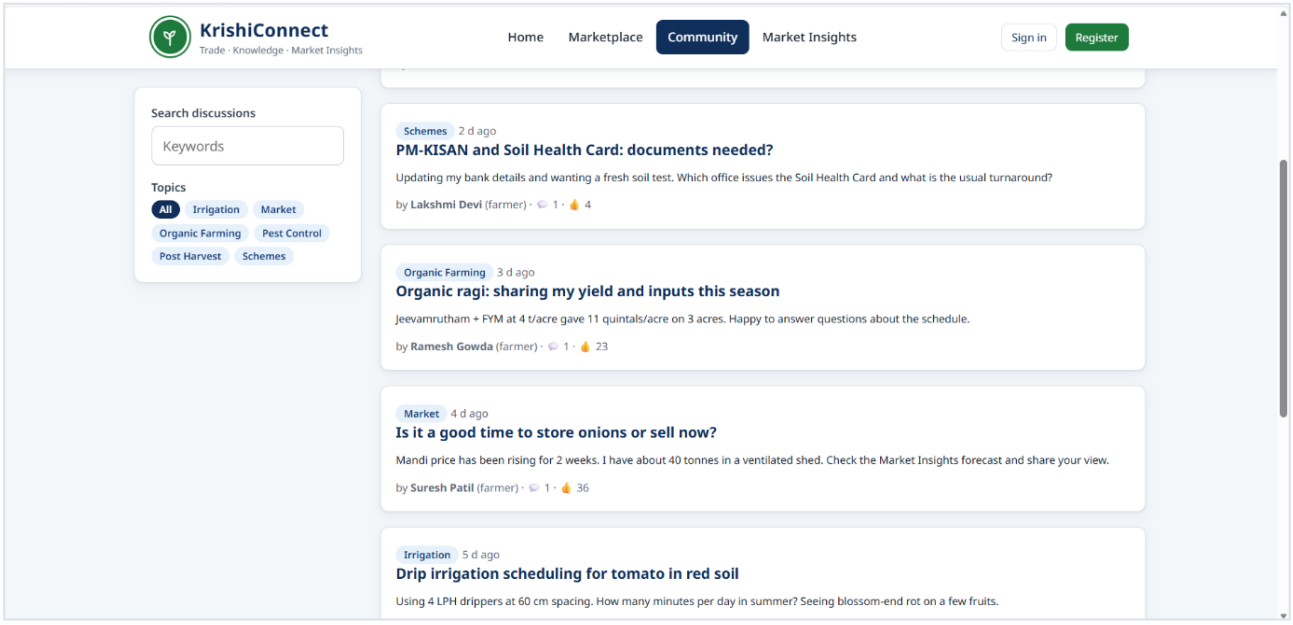

Users choose a commodity, the history window (90 days here) and the forecast horizon (14 days). Summary cards show the latest modal price, the period high and low, and the forecast change. For groundnut: latest 6,027.59, high 7,346.48, low 5,686.12 (Rs per quintal), forecast +1%.

## Figure 5. Market Insights - price chart and forecast



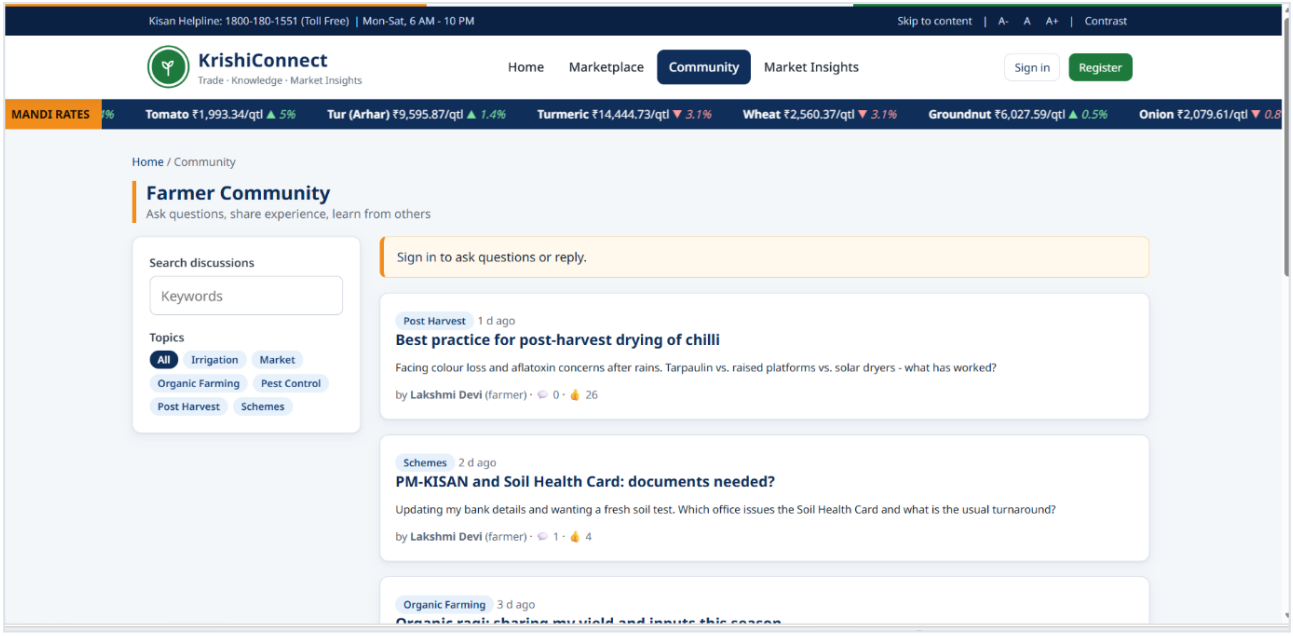

The chart overlays the actual price, its 7-day average and the ARIMA forecast (dashed, in the shaded region). The outlook note reports the fitted model, ARIMA(2,1,2): prices are expected to remain stable, with an expected price of Rs 6,087.41 on 15 Nov.

## Figure 6. Farmer Community - topics




The community page lets users search discussions by keyword and filter by topic: Irrigation, Market, Organic Farming, Pest Control, Post Harvest and Schemes. Each post shows its topic, age, title, an excerpt, the author and role, reply count and likes.

## Figure 7. Farmer Community - signed-out view




Visitors who are not signed in can read discussions, and a notice invites them to sign in to ask questions or reply. Posts such as post-harvest drying of chilli and PM-KISAN / Soil Health Card documents illustrate the knowledge-sharing use case.

# Summary

The screenshots show the main workflows of KrishiConnect end to end: discovering produce, listing a product, reading price trends with a statistical forecast, and learning through the community. Role-based navigation changes what each user sees, and accessibility controls are available on every page.